

Demonstration of Proposed Features

Project: Rising Water – AI-Based Flood Prediction System

1. Demonstration Objective

Demonstrate how the Rising Water application predicts flood probability using Machine Learning and a Flask web interface.

2. Features Demonstrated

Feature	Description
User-Friendly Interface	Simple web form for entering 20 environmental parameters.
Input Validation	Accepts only valid input values.
ML Prediction	Uses the trained XGBoost model to predict flood probability.
Risk Classification	Displays Low, Medium, or High risk.
Fast Response	Prediction generated within a few seconds.

3. Demonstration Steps

1. Launch the Flask application.
2. Open the application in a browser.
3. Enter all required parameters.
4. Click 'Predict Flood Risk'.
5. Observe the predicted probability and risk level.

4. Expected Outcome

The application successfully predicts flood probability and displays the corresponding risk category, demonstrating the practical use of Machine Learning for disaster preparedness.

5. Conclusion

The demonstration confirms that the Rising Water system is functional, easy to use, and capable of providing quick flood risk assessments.